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Mental Health — Burnout, Stress and Sedentary Behavior Among Computer Science students

A Descriptive Statistical Report

Nahid Hasan Rayan | Group Boleh | Section 15

10 May 2026

1 Introduction

Computing students experience some of the highest stress levels among undergraduates, yet their daily habits are rarely examined with the depth they deserve. Late nights, long hours in front of screens, and constant exhaustion have become so common that we hardly notice them.

This study started from a personal observation. As computing students ourselves, I have seen classmates sacrifice sleep for deadlines, skip meals while debugging, and talk about burnout as if it were just part of the degree. we wanted to know whether these experiences are genuinely widespread or only felt by a few. The data, as you will see, confirm that they are widespread and in some areas, more severe than we expected.

We focused on undergraduates in computing-related programmes across Malaysian universities. A structured survey captured daily screen time, sleep, physical activity, sitting hours, bedtime, and academic exhaustion. Perceived stress was measured with the **Perceived Stress Scale (PSS-10)** developed by Cohen, Kamarck and Mermelstein (1983), a well validated instrument used in over 40 countries.

Using a validated instrument matters. The PSS-10 gives us two things that informal questions cannot: a standardised score on a known scale, and published population norms against which we can compare our results. That comparison is where the most striking finding emerges.

Before collecting any data, we already expected high screen time, little sleep, minimal exercise, and stress levels far above what anyone would consider sustainable. The analysis confirmed every one of those expectations and the sleep deprivation and PSS-10 scores were even more striking than we had anticipated.

2 Data Collection

2.1 Survey Instrument and Sampling

The survey ran as an Google Form from March to April 2026. Participants were recruited through WhatsApp, Discord, Telegram, and classroom outreach a convenience sampling and Snowball Sampling approach drawing on our academic and social networks.

The questionnaire contained **27 structured items** across four themes:

Survey structure

Block	Theme	Variables Captured	Item Count
Block 1	**Demographics**	Gender, age, university, year of study, programme, CGPA	6 items
Block 2	**Lifestyle Habits**	Daily screen time, sleep duration, bedtime, sitting hours, exercise frequency and session duration	6 items
Block 3	**Academic Wellbeing**	Exhaustion frequency, days absent due to mental fatigue, workload engagement state	3 items
Block 4	**Perceived Stress (PSS-10)**	All 10 items of Cohen et al.'s (1983) validated scale with standard 5-point Likert response options	10 items + 1 open-ended

Informed consent was on the first page. Participation was voluntary and anonymous; no personally identifying information was kept.

2.2 Data Cleaning in R

The raw CSV was cleaned programmatically in R before analysis. Every step is documented below and reproducible from Appendix B.

Data cleaning pipeline (fully reproducible – see Appendix B)

Step	Action	Rationale
Step 1	Removed the participant name column and the open-ended closing question	Name field is personally identifying and cannot be used in statistical analysis
Step 2	Renamed all 26 remaining variables to R-friendly identifiers (e.g., ScreenHours, SleepHours)	Prevents R parsing errors and improves code readability across the entire pipeline
Step 3	Recoded PSS-10 text responses to numeric values (Never = 0 through Very Often = 4)	Required for score computation per the original PSS-10 scoring protocol
Step 4	Reverse-scored PSS items 4, 5, 7, and 8 using the formula: reversed = 4 minus original	Items 4, 5, 7, 8 are positively worded; failing to reverse them would invalidate all PSS-10 results
Step 5	Computed each participant's PSS-10 total score (valid range: 0-40)	Produces a continuous 0-40 stress index for each respondent
Step 6	Converted all categorical variables to ordered or unordered factor types in R	Ensures correct handling in frequency tables, charts, and all downstream summaries
Step 7	Standardised university names to uppercase and trimmed leading or trailing whitespace	Resolves capitalisation inconsistencies that would otherwise create spurious duplicate categories
Step 8	Removed logically impossible values (e.g., SleepHours > 24, ScreenHours > 24)	Data-entry errors or nonsense values in continuous fields would bias all central tendency measures
Step 9	Filtered dataset to computing-related programmes only	Maintains conceptual coherence – this study concerns computing students specifically
Step 10	Retained only complete cases on Age, SleepHours, and PSS_total	Preserves the integrity of the three key analytical variables used throughout

Why reverse-scoring matters. The PSS-10 contains four positively worded items for example, "How often have you felt confident about your ability to handle your personal problems?" A high answer on these items signals *low* stress, not high stress. Summing without reversing would corrupt the total score. Every total score in this report uses correctly reverse-scored items 4, 5, 7, and 8.

2.3 Final Sample

Final sample overview

Characteristic	Value
Total respondents (after cleaning)	n = 98
Male respondents	60 (61.2%)
Female respondents	38 (38.8%)
Mean age (SD)	20.6 years (SD = 1.6)
Universities represented	7 institutions
Minimum required by rubric	**n = 60 – satisfied**

Sampling caveat. This study used convenience sampling through personal networks, which brings a well-known self-selection bias, respondents tend to share characteristics with the researchers. Also, over 69% of respondents are first-year students. These two features shape the interpretation of every result, and we return to them throughout the analysis.

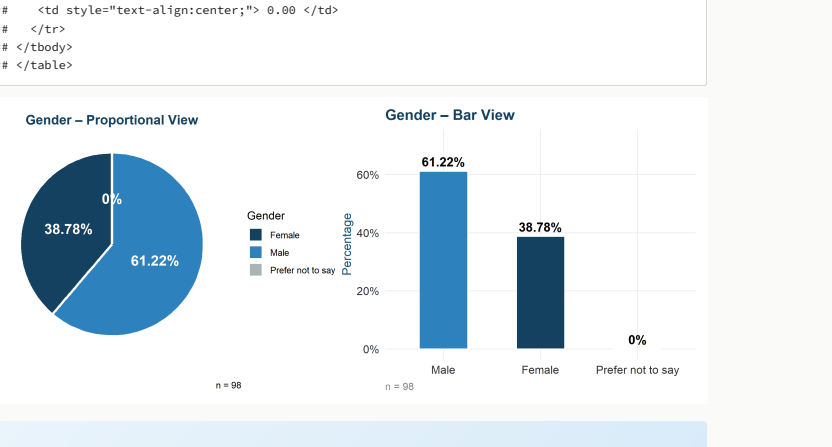
3 Data Analysis

3.1 Categorical Variables

For each categorical variable we show a frequency table and the most appropriate chart. Pie charts are used for few-category variables, bar charts for ordered or multi-category variables. All percentages use n = 98.

3.1.1 Gender

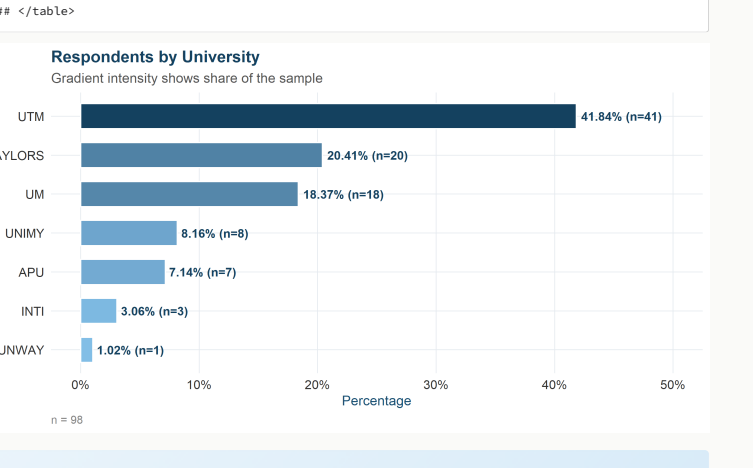
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution of Gender</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> Male </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 60 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 61.22 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Female </td>
##    <td style="text-align:center;"> 38 </td>
##    <td style="text-align:center;"> 38.78 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Prefer not to say </td>
##    <td style="text-align:center;"> 0 </td>
##    <td style="text-align:center;"> 0.00 </td>
##   </tr>
## </tbody>
## </table>
```



What the data shows. The gender split – 61.22% male, 38.78% female – is less skewed than the global computing average. UNESCO (2021) reports about 30% female participation in computing degrees in Southeast Asia. Cheryan et al. (2017) found that female students in male-majority technical fields report higher belonging uncertainty, which independently raises perceived stress. Our aggregate PSS-10 scores may hide this within-gender variation, and separating by gender is an important next step.

3.1.2 University

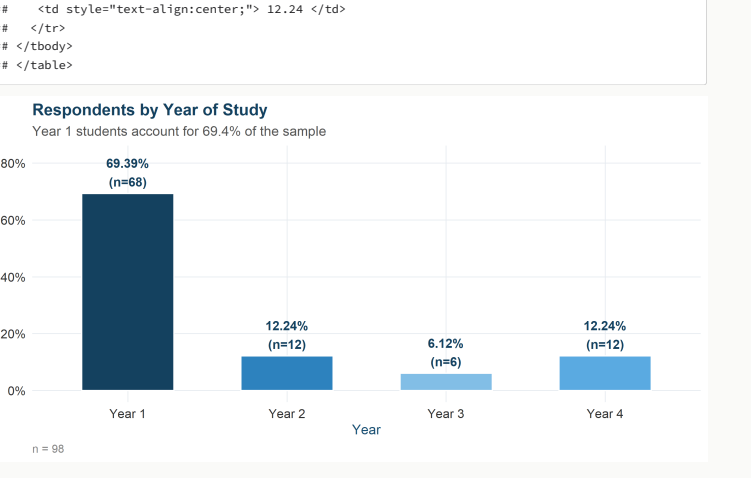
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution by University</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;">  </th>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 7 </td>
##    <td style="text-align:left;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> UTM </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 41 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 41.84 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 4 </td>
##    <td style="text-align:left;"> TAYLORS </td>
##    <td style="text-align:center;"> 20 </td>
##    <td style="text-align:center;"> 20.41 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 5 </td>
##    <td style="text-align:left;"> UM </td>
##    <td style="text-align:center;"> 18 </td>
##    <td style="text-align:center;"> 18.37 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 6 </td>
##    <td style="text-align:left;"> UNIMY </td>
##    <td style="text-align:center;"> 8 </td>
##    <td style="text-align:center;"> 8.16 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 1 </td>
##    <td style="text-align:left;"> APU </td>
##    <td style="text-align:center;"> 7 </td>
##    <td style="text-align:center;"> 7.14 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 2 </td>
##    <td style="text-align:left;"> INTI </td>
##    <td style="text-align:center;"> 3 </td>
##    <td style="text-align:center;"> 3.06 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 3 </td>
##    <td style="text-align:left;"> SUNWAY </td>
##    <td style="text-align:center;"> 1 </td>
##    <td style="text-align:center;"> 1.02 </td>
##   </tr>
## </tbody>
## </table>
```



What the data shows. UTM contributes the largest share, followed by Taylors and UM. This reflects our outreach channels rather than enrolment proportions. Responses came from 7 distinct institutions, giving the dataset a breadth that a single-campus study would lack, although the findings cannot be claimed as representative of any one institution or of all Malaysian computing students.

3.1.3 Year of Study

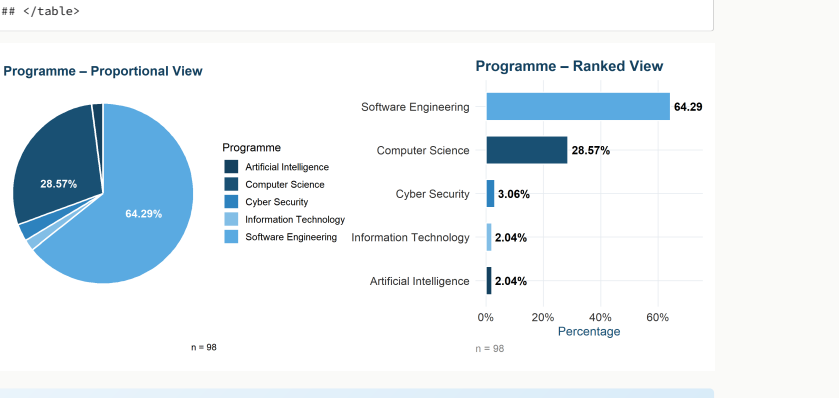
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution by Year of Study</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> Year 1 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 68 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 69.39 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Year 2 </td>
##    <td style="text-align:center;"> 12 </td>
##    <td style="text-align:center;"> 12.24 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Year 3 </td>
##    <td style="text-align:center;"> 6 </td>
##    <td style="text-align:center;"> 6.12 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Year 4 </td>
##    <td style="text-align:center;"> 12 </td>
##    <td style="text-align:center;"> 12.24 </td>
##   </tr>
## </tbody>
## </table>
```



A critical contextual factor. Year 1 students make up **69.4%** of this sample. The first year of university is one of the most psychologically demanding life transitions, simultaneous adjustments in academic demands, social life, independence, and identity happen in a short time with few established coping resources (Cooke et al., 2006). The stress and exhaustion indicators throughout this report should be read with that in mind. They likely reflect both the acute first year adjustment and the longer term reality of computing education.

3.1.4 Programme

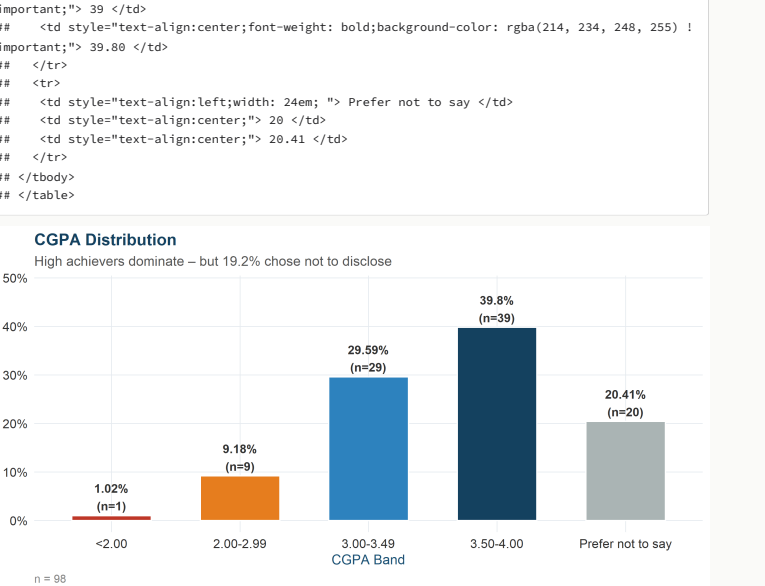
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution by Programme</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Artificial Intelligence </td>
##    <td style="text-align:center;"> 2 </td>
##    <td style="text-align:center;"> 2.04 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Computer Science </td>
##    <td style="text-align:center;"> 28 </td>
##    <td style="text-align:center;"> 28.57 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Cyber Security </td>
##    <td style="text-align:center;"> 3 </td>
##    <td style="text-align:center;"> 3.06 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Information Technology </td>
##    <td style="text-align:center;"> 2 </td>
##    <td style="text-align:center;"> 2.04 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> Software Engineering </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 63 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 64.29 </td>
##   </tr>
## </tbody>
## </table>
```



What the data show. Software Engineering makes up 64.3% of the sample and Computer Science another 28.6%. The remaining respondents are from Cyber Security, Artificial Intelligence, and Information Technology. All five programmes share the core features relevant to this study: screen intensive, cognitively demanding coursework with regular deadline pressure, so they form a coherent analysis group.

3.1.5 CGPA

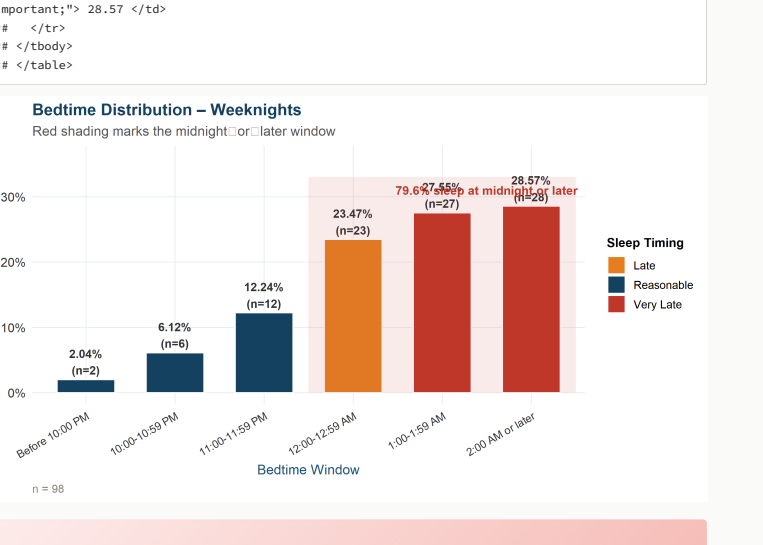
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution by CGPA Band</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; "> &lt;2.00 </td>
##    <td style="text-align:center;"> 1 </td>
##    <td style="text-align:center;"> 1.02 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 2.00–2.99 </td>
##    <td style="text-align:center;"> 9 </td>
##    <td style="text-align:center;"> 9.18 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 3.00–3.49 </td>
##    <td style="text-align:center;"> 29 </td>
##    <td style="text-align:center;"> 29.59 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 3.50–4.00 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 39 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 39.80 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Prefer not to say </td>
##    <td style="text-align:center;"> 20 </td>
##    <td style="text-align:center;"> 20.41 </td>
##   </tr>
## </tbody>
## </table>
```



What the data shows. The concentration at higher CGPA bands likely comes from two things: real high achievement and self-selection, where high achievers feel more comfortable taking part in academic surveys. The 20.4% who chose not to share their CGPA is itself informative. In a setting where academic results affect scholarships, family expectations, and career prospects, non-disclosure often signals grade-related anxiety rather than simple privacy preference.

3.1.6 Bedtime

```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution of Bedtime (Weeknights)</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Before 10:00 PM </td>
##    <td style="text-align:center;"> 2 </td>
##    <td style="text-align:center;"> 2.04 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 10:00–10:59 PM </td>
##    <td style="text-align:center;"> 6 </td>
##    <td style="text-align:center;"> 6.12 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 11:00–11:59 PM </td>
##    <td style="text-align:center;"> 12 </td>
##    <td style="text-align:center;"> 12.24 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 12:00–12:59 AM </td>
##    <td style="text-align:center;"> 23 </td>
##    <td style="text-align:center;"> 23.47 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 1:00–1:59 AM </td>
##    <td style="text-align:center;"> 27 </td>
##    <td style="text-align:center;"> 27.55 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 2:00 AM or later </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 28 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 28.57 </td>
##   </tr>
## </tbody>
## </table>
```



A deeply concerning pattern. Only 20.4% are in bed by 11:00 PM. A striking **28.6%** are routinely awake past 2:00 AM, and **79.6%** go to bed at midnight or later. Sleep timing is not just a lifestyle choice, it is a physiological parameter. Sustained circadian misalignment of this magnitude is linked to impaired executive function, reduced working memory, and significantly elevated perceived stress, independent of total sleep quantity (Killgore, 2010).

3.1.7 Exercise Frequency

```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Exercise Frequency per Week</caption>
##  <thead>
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##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 0 – I do not exercise </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 38 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 38.78 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 1 time </td>
##    <td style="text-align:center;"> 11 </td>
##    <td style="text-align:center;"> 11.22 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 2–3 times </td>
##    <td style="text-align:center;"> 36 </td>
##    <td style="text-align:center;"> 36.73 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> 4–5 times </td>
##    <td style="text-align:center;"> 13 </td>
##    <td style="text-align:center;"> 13.27 </td>
##   </tr>
## </tbody>
## </table>
```
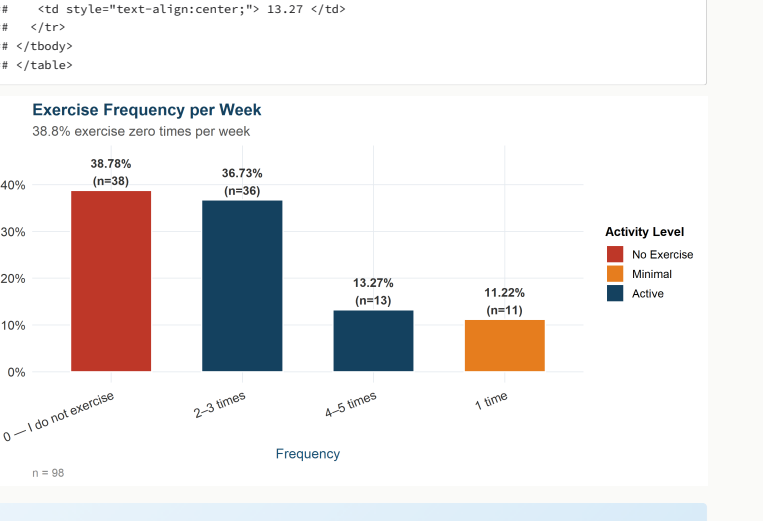
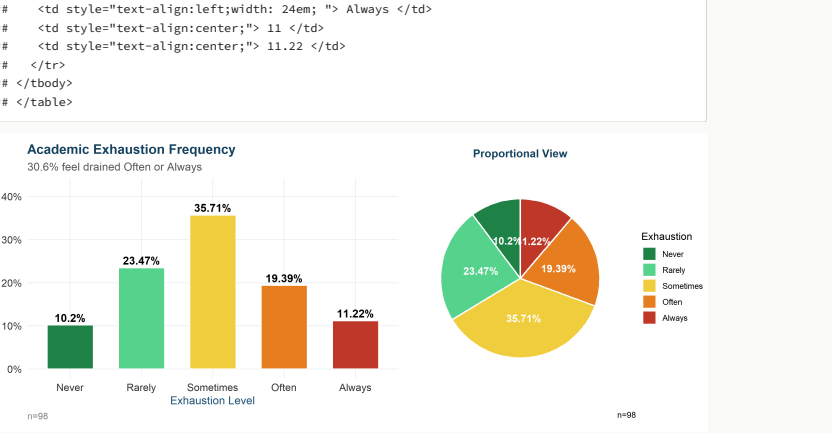



What the research says. A meta-analysis by Rebar et al. (2015), covering 40 studies, found significant reductions in perceived stress following aerobic exercise programmes of only eight weeks. With **38.8%** never exercising at all, and a further 11.2% exercising once a week, nearly half the sample lacks access to one of the most reliable psychological resilience tools, precisely when their stress is highest.

3.1.8 Academic Exhaustion

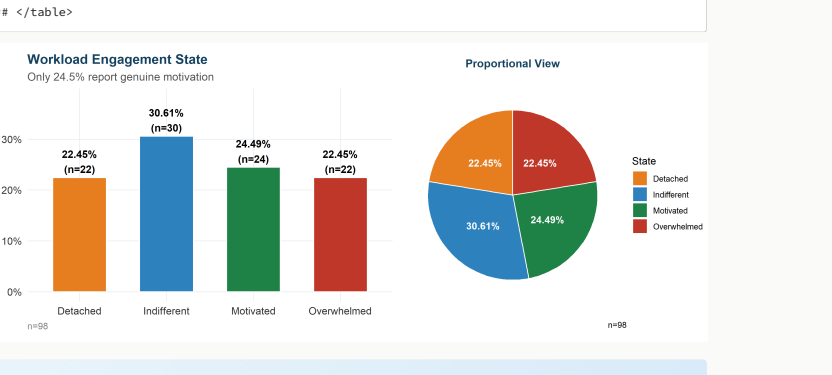
```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Distribution of Academic Exhaustion</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Never </td>
##    <td style="text-align:center;"> 10 </td>
##    <td style="text-align:center;"> 10.20 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Rarely </td>
##    <td style="text-align:center;"> 23 </td>
##    <td style="text-align:center;"> 23.47 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> Sometimes </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 35 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 35.71 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Often </td>
##    <td style="text-align:center;"> 19 </td>
##    <td style="text-align:center;"> 19.39 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Always </td>
##    <td style="text-align:center;"> 11 </td>
##    <td style="text-align:center;"> 11.22 </td>
##   </tr>
## </tbody>
## </table>
```



What the research says. Academic exhaustion is the main precursor to full clinical burnout (Maslach & Leiter, 2016). In our sample, **30.6%** of students report this as their regular experience. A further 35.7% experience it sometimes. Fewer than one in eight respondents report never feeling exhausted sustained depletion has become the normal state for this cohort.

3.1.9 Workload Engagement

```
## <table class="table table-striped table-hover table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
## <caption style="font-size: initial !important;">Workload Engagement State</caption>
##  <thead>
##   <tr>
##    <th style="text-align:left;"> Category </th>
##    <th style="text-align:center;"> n </th>
##    <th style="text-align:center;"> % </th>
##   </tr>
##  </thead>
## <tbody>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Motivated </td>
##    <td style="text-align:center;"> 24 </td>
##    <td style="text-align:center;"> 24.49 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> Indifferent </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 30 </td>
##    <td style="text-align:center;font-weight: bold;background-color: rgba(214, 234, 248, 255) !important;"> 30.61 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Detached </td>
##    <td style="text-align:center;"> 22 </td>
##    <td style="text-align:center;"> 22.45 </td>
##   </tr>
##   <tr>
##    <td style="text-align:left;width: 24em; "> Overwhelmed </td>
##    <td style="text-align:center;"> 22 </td>
##    <td style="text-align:center;"> 22.45 </td>
##   </tr>
## </tbody>
## </table>
```



What the data reveals. Only **24.5%** describe themselves as genuinely engaged and in control. The remaining 75.5% fall between passive compliance and active overwhelm. Compliance and genuine engagement produce very different educational outcomes. A group where fewer than one in four feels motivated is one whose intellectual potential is being substantially underrealised, and that has implications for students and for the field that trains them.

3.2 Quantitative Variables

3.2.1 Summary Statistics

Summary Statistics – All Quantitative Variables

	Central Tendency			Dispersion			
	Mean	**Median**	**Mode**	**SD**	**Min**	**Max**	**IQR**
Age (years)	20.63	20.0	20	1.62	18	25	1.00
Sitting Hours / day	6.77	6.0	5	3.54	0	24	3.00
Screen Hours / day	9.04	9.0	8	2.90	1	15	5.00
Sleep Hours / night	6.48	7.0	7	1.74	2	12	2.75
Exercise Hours / session	0.78	1.0	1	0.72	0	2	1.00
Days Off (mental fatigue)	4.39	1.5	0	7.50	0	30	5.00
PSS-10 Total Score	18.89	19.0	17	3.77	5	30	4.00

9h MEAN SCREEN / DAY

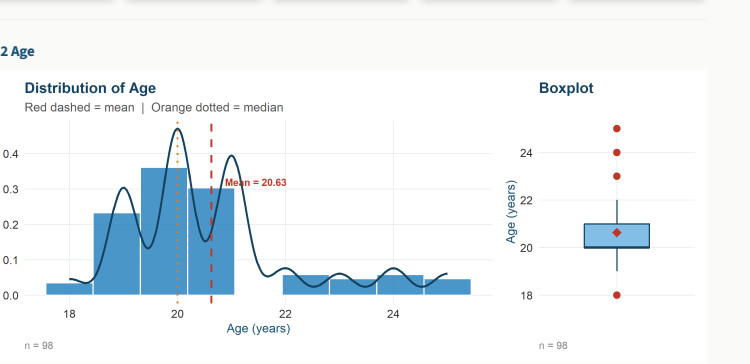
6.5h MEAN SLEEP / NIGHT

6.8h MEAN SITTING / DAY

38.8% NEVER EXERCISE

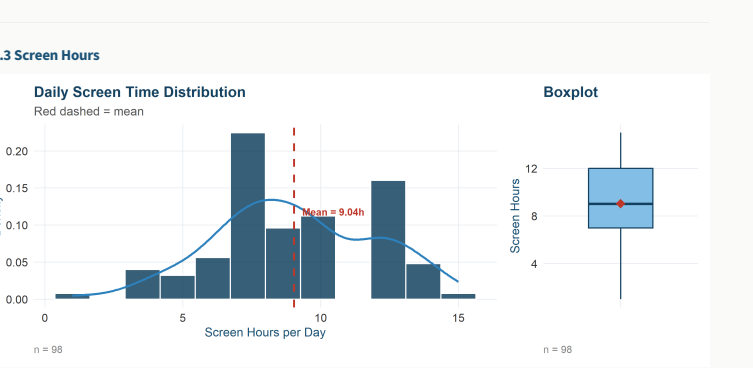
18.9 MEAN PSS-10 SCORE

3.2.2 Age



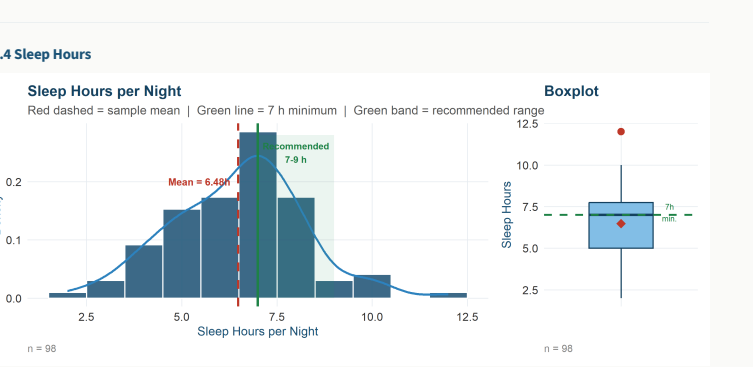
Age is tightly clustered around 20.63 years (SD = 1.62). The near-identical mean and median confirm a symmetric distribution. This is a typical undergraduate cohort aged 18–23 a period of both high cognitive flexibility and heightened vulnerability to stress-related mental health disruption (Arnett, 2007).

3.2.3 Screen Hours



A mean of **9.04 hours** with an SD of only 2.9 shows that high screen time is the norm, not an outlier feature. For computing students, screen time is not only recreational; programming environments, lecture recordings, and collaborative tools like GitHub all require it. The real question is how screen-based study can be structured to include regular physical breaks, and what role course design plays.

3.2.4 Sleep Hours



3.2.4.1 Stem-and-Leaf Plot – Sleep Hours

```
Stem-and-Leaf: Sleep Hours per Night
```

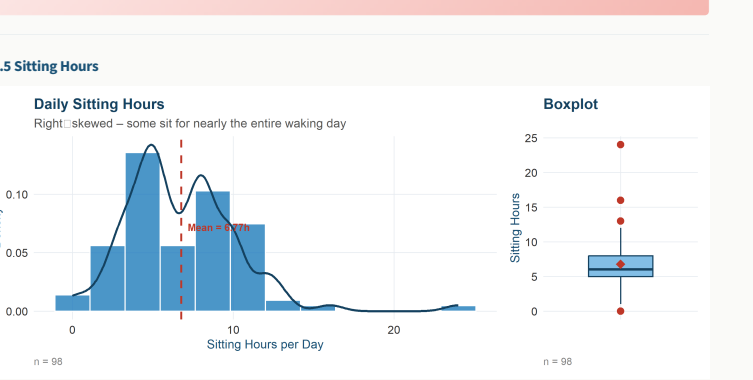
```
Each leaf = one respondent  |  Decimal point at the |
```

```
  The decimal point is at the |

   2 | 0000
   4 | 000000000000000000000
   6 | 000000000000000000000000000000000000000000000000
   8 | 00000000000000000000
  10 | 0000
  12 | 0
```

```
Key: 6 | 5  reads as 6.5 hours of sleep
```

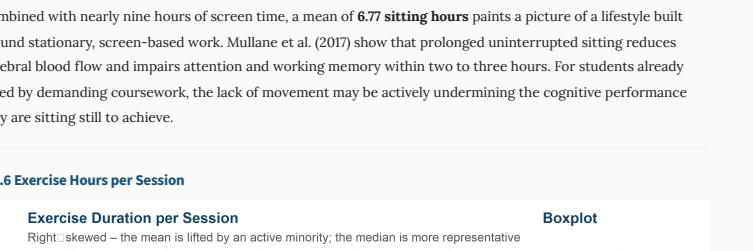
The sleep finding. A mean of **6.48 hours** falls below the 7-hour minimum recommended for healthy adult cognition. The IQR of 2.75 hours, centred on a median of 7, means that the bottom 25% are sleeping roughly 5.5 hours or less. Walker (2017) shows that 17-19 hours of wakefulness produces cognitive impairment comparable to a blood alcohol level above the legal driving limit. For students going to bed at 2:00 AM and waking for a 9:00 AM lecture, that threshold is crossed every single day.

3.2.5 Sitting Hours



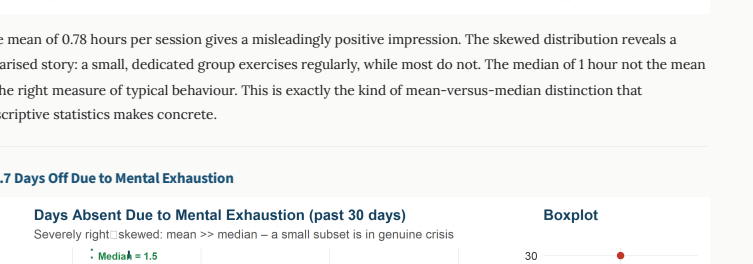
Combined with nearly nine hours of screen time, a mean of **6.77 sitting hours** paints a picture of a lifestyle built around stationary, screen-based work. Mullane et al. (2017) show that prolonged uninterrupted sitting reduces cerebral blood flow and impairs attention and working memory within two to three hours. For students already taxed by demanding coursework, the lack of movement may be actively undermining the cognitive performance they are sitting still to achieve.

3.2.6 Exercise Hours per Session



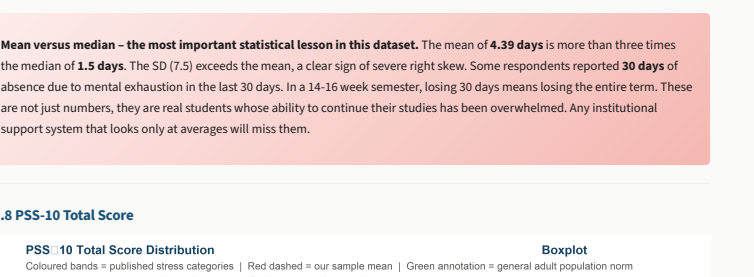
The mean of 0.78 hours per session gives a misleadingly positive impression. The skewed distribution reveals a polarised story: a small, dedicated group exercises regularly, while most do not. The median of 1 hour not the mean is the right measure of typical behaviour. This is exactly the kind of mean-versus-median distinction that descriptive statistics makes concrete.

3.2.7 Days Off Due to Mental Exhaustion



Mean versus median – the most important statistical lesson in this dataset. The mean of **4.39 days** is more than three times the median of **1.5 days**. The SD (7.5) exceeds the mean, a clear sign of severe right skew. Some respondents reported **30 days** of absence due to mental exhaustion in the last 30 days. In a 14-16 week semester, losing 30 days means losing the entire term. These are not just numbers, they are real students whose ability to continue their studies has been overwhelmed. Any institutional support system that looks only at averages will miss them.

3.2.8 PSS-10 Total Score

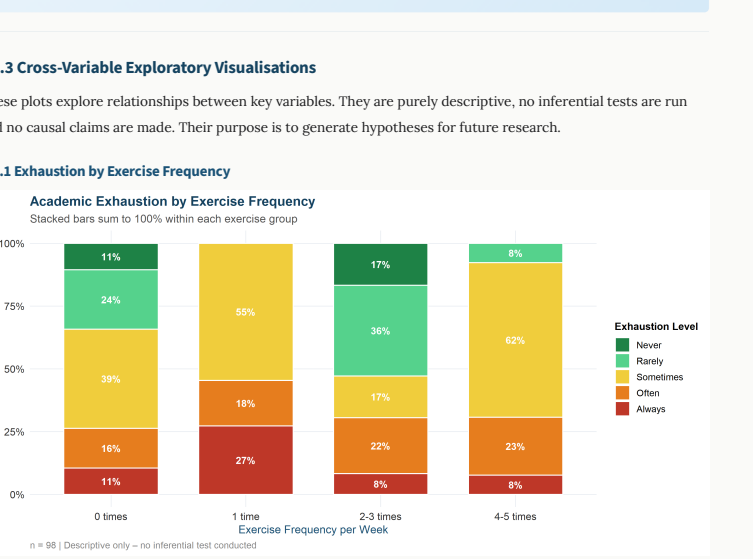


Benchmarking against published norms. The PSS-10 is scored 0-40. Cohen and Williamson's (1988) reference study on a nationally representative US adult sample reported a mean of about **13.0**. Our mean of **18.89** is **5.9 points above** that baseline – more than one standard deviation above the population norm. The IQR of 4 confirms this elevation is widely shared, not driven by a few extreme cases. This group is not experiencing ordinary background stress. They are experiencing stress measurably and meaningfully above general population norms, at a level that in clinical contexts would warrant proactive attention.

3.3 Cross-Variable Exploratory Visualisations

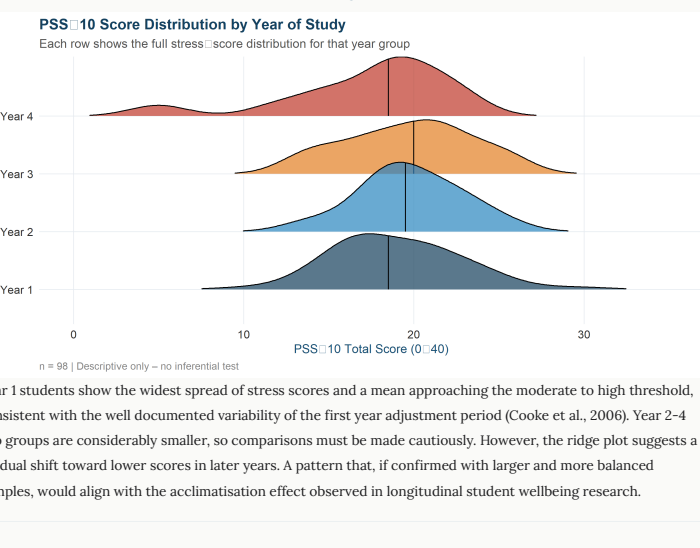
These plots explore relationships between key variables. They are purely descriptive, no inferential tests are run and no causal claims are made. Their purpose is to generate hypotheses for future research.

3.3.1 Exhaustion by Exercise Frequency



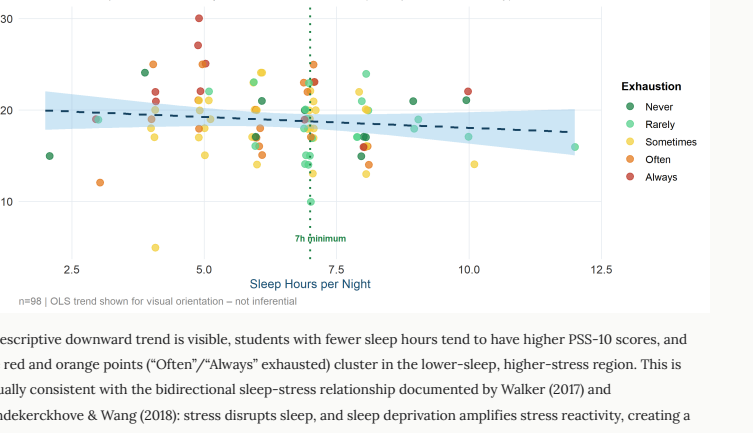
Among students who never exercise, the "Often" and "Always" exhaustion categories occupy a visibly larger combined share compared to those exercising four to five times weekly, where "Never" and "Rarely" exhaustion are proportionally more prominent. This pattern is consistent with the literature on physical activity as a buffer against psychological fatigue (Rebar et al., 2015) and forms a clear hypothesis for future testing with larger samples and formal statistical methods.

3.3.2 PSS-10 Score by Year of Study (Colourful Density Ridges)



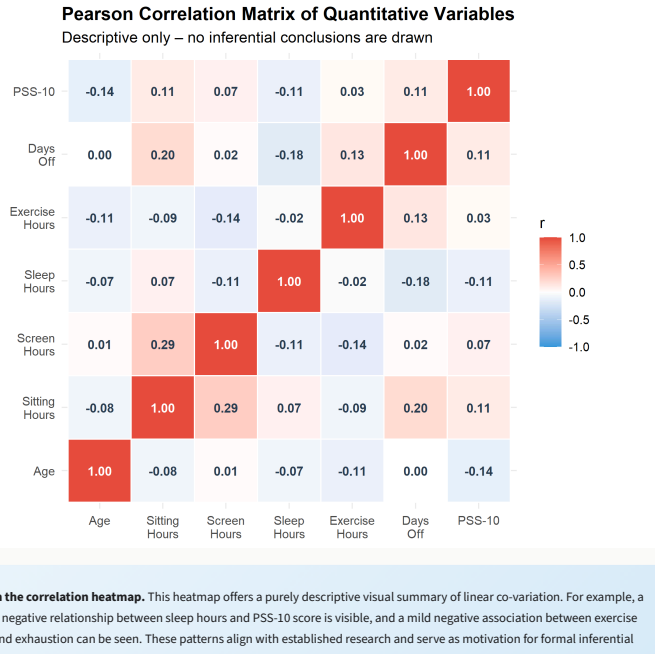
Year 1 students show the widest spread of stress scores and a mean approaching the moderate to high threshold, consistent with the well documented variability of the first year adjustment period (Cooke et al., 2006). Year 2–4 sub groups are considerably smaller, so comparisons must be made cautiously. However, the ridge plot suggests a gradual shift toward lower scores in later years. A pattern that, if confirmed with larger and more balanced samples, would align with the acclimatisation effect observed in longitudinal student wellbeing research.

3.3.3 Sleep Hours vs. PSS-10 Score



A descriptive downward trend is visible, students with fewer sleep hours tend to have higher PSS-10 scores, and the red and orange points ("Often"/"Always" exhausted) cluster in the lower-sleep, higher-stress region. This is visually consistent with the bidirectional sleep-stress relationship documented by Walker (2017) and Vandekerckhove & Wang (2018): stress disrupts sleep, and sleep deprivation amplifies stress reactivity, creating a self-reinforcing cycle that sustains burnout across the semester.

3.3.4 Correlation Heatmap of Quantitative Variables



Note on the correlation heatmap. This heatmap offers a purely descriptive visual summary of linear co-variation. For example, a modest negative relationship between sleep hours and PSS-10 score is visible, and a mild negative association between exercise hours and exhaustion can be seen. These patterns align with established research and serve as motivation for formal inferential testing in future phases. No causal conclusions are drawn here.

4 Conclusion

This study set out to describe, with empirical precision, the mental health and lifestyle patterns of computing undergraduates across Malaysian universities. What we found was internally consistent and, in the most important respects, concerning.

Summary of Key Findings

Indicator	Finding	Signal
Daily screen time	Mean 9h – the dominant feature of the waking day	Elevated
Sleep per night	Mean 6.5h – below the 7h health threshold; 83% sleep past midnight	Critical
Bedtime	79.6% go to bed at midnight or later; 28.6% past 2:00 AM	Critical
Physical activity	38.8% report zero exercise per week	Critical
Academic exhaustion	30.6% feel drained Often or Always	Elevated
Workload engagement	Only 24.5% describe themselves as genuinely motivated	Elevated
Perceived stress (PSS-10)	Mean 18.9 – approximately 5.9 points above published population norms	Critical
Days lost to exhaustion	Mean 4.4 days (range 0-30); median is only 1.5 days	Critical

What this study taught us about statistics. The most durable methodological lesson is that no single summary statistic tells the full story. The "Days Off" variable makes this powerfully clear: a mean of 4.4 days, presented alone, would suggest modest disruption. The median of 1.5 days, the mode of 0, and the maximum of 30, read together, reveal a severely skewed distribution where most students persist while a small subset has been effectively lost to mental health difficulties. An analyst who reports only the mean misrepresents the data. Triangulating across mean, median, SD, IQR, and visual tools is not statistical formality, it is what honesty requires.

The PSS-10 results gave us the second major lesson: numbers need reference points to mean anything. A mean of 18.9 is unremarkable in isolation. Benchmarked against Cohen and Williamson's (1988) general adult norm of about 13.0, it becomes a finding. Statistics without context are noise; statistics anchored to established norms are evidence.

What this study taught us technologically. Building the full analysis in R Markdown from raw CSV ingestion through every cleaning step, summary table, and visualisation was an exercise in the reproducible research practices used by professional data organisations. Every number and every chart is generated live from the source data. Correcting a cleaning error propagates automatically through the entire document on re-knitting. We also developed practical fluency with `dplyr`, `ggplot2`, `patchwork`, and `kableExtra` – a toolkit that transfers directly to professional analytical environments.

Limitations and the path forward. With n = 98 respondents drawn through convenience/snowfall sampling and weighted toward first-year students at UTM and Taylors, these findings cannot be generalised to all Malaysian computing students. The study is strictly descriptive, no causal conclusions are drawn or warranted. A more rigorous follow up would pursue stratified random sampling across year groups and institutions, a longitudinal design tracking stress across a full academic year, inferential statistical testing of the associations suggested here, and a qualitative component structured interviews to explain the mechanisms that survey data alone cannot reveal.

What this study provides is a precise descriptive baseline. These are the numbers, drawn from real students who gave their time to answer 27 questions about how they are actually doing. They are now documented with specificity, with methodological care, and in a form that can be shared, cited, and acted upon.

5 References

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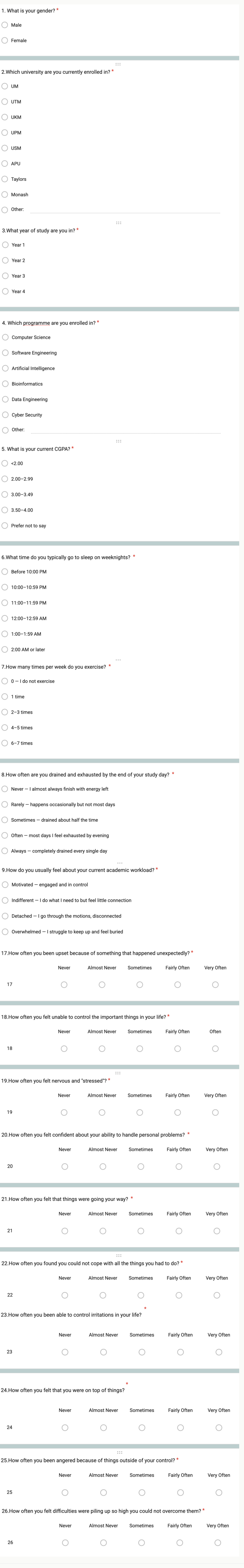
6 Appendix A: Questionnaire



1. What is your gender? *
○ Male
○ Female

2.Which university are you currently enrolled in? *
○ UM
○ UTM
○ UKM
○ UPM
○ USM
○ APU
○ Taylors
○ Monash
○ Other:

3.What year of study are you in? *
○ Year 1
○ Year 2
○ Year 3
○ Year 4

4. Which programme are you enrolled in? *
○ Computer Science
○ Software Engineering
○ Artificial Intelligence
○ Bioinformatics
○ Data Engineering
○ Cyber Security
○ Other:

5. What is your current CGPA? *
○ <2.00
○ 2.00–2.99
○ 3.00–3.49
○ 3.50–4.00
○ Prefer not to say

6.What time do you typically go to sleep on weeknights? *
○ Before 10:00 PM
○ 10:00–10:59 PM
○ 11:00–11:59 PM
○ 12:00–12:59 AM
○ 1:00–1:59 AM
○ 2:00 AM or later

7.How many times per week do you exercise? *
○ 0 — I do not exercise
○ 1 time
○ 2–3 times
○ 4–5 times
○ 6–7 times

8.How often are you drained and exhausted by the end of your study day? *
○ Never — I almost always finish with energy left
○ Rarely — happens occasionally but not most days
○ Sometimes — drained about half the time
○ Often — most days I feel exhausted by evening
○ Always — completely drained every single day

9.How do you usually feel about your current academic workload? *
○ Motivated — engaged and in control
○ Indifferent — I do what I need to but feel little connection
○ Detached — I go through the motions, disconnected
○ Overwhelmed — I struggle to keep up and feel buried

17.How often you been upset because of something that happened unexpectedly? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
17	○	○	○	○	○

18.How often you felt unable to control the important things in your life? *

	Never	Almost Never	Sometimes	Fairly Often	Often
18	○	○	○	○	○

19.How often you felt nervous and "stressed"? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
19	○	○	○	○	○

20.How often you felt confident about your ability to handle personal problems? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
20	○	○	○	○	○

21.How often you felt that things were going your way? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
21	○	○	○	○	○

22.How often you found you could not cope with all the things you had to do? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
22	○	○	○	○	○

23.How often you been able to control irritations in your life? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
23	○	○	○	○	○

24.How often you felt that you were on top of things? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
24	○	○	○	○	○

25.How often you been angered because of things outside of your control? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
25	○	○	○	○	○

26.How often you felt difficulties were piling up so high you could not overcome them? *

	Never	Almost Never	Sometimes	Fairly Often	Very Often
26	○	○	○	○	○

7 Appendix B: Complete R Code

Every step of the analysis data loading, cleaning, variable recoding, summary statistics, and all visualisations is embedded in this R Markdown source file. To reproduce the full report: open `Project1_GroupBoleh.Rmd` in RStudio and choose **Knit to HTML**. The entire pipeline runs from raw data to finished document in a single command.

Submitted in partial fulfilment of the requirements for SECP1223: Probability and Statistical Data Analysis, Section 15 – Academic Year 2025/2026, Semester 2.